LECTURE ACTIVITY 1: STATISTICAL STORYTELLING – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

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Learning Objectives:

By the end of this activity, you should be able to

1) Define and identify observational units.
2) Define and identify variables.
3) Classify variables as categorical or quantitative.
4) Extract basic information from simple data visualizations.
5) Interact with and manipulate variables within plots.

Welcome!

Welcome to our first Lecture Activity! To embody Cal Poly's spirit of "Learn by Doing," we'll frequently (almost every lecture) incorporate a Lecture Activity like this one that is designed to give you a more hands-on approach to the course material than a "standard" lecture would. I (Dr. Ethan) highly encourage you to work in groups, as statistics is meant to be a very collaborative field!

Part 1: A Video

Watch the video found at the following link (for those of you who printed out this activity, here is the url: `https://bit.ly/1110-la01`). We'll use the plot in this video to answer a few questions.

Question 1

What's the "story" this plot tells us? (This is an intentionally open-ended question!)

Solution: Answers may vary! Generally speaking, there are a few things we can see from the video:

- Countries with higher GDPs tend to have higher average life expectancies than countries with lower GDPs. This is indicated by the fact that the points for each plot fall roughly along a line with a positive slope - i.e. as the x-coordinates increase, so too (on average) do the y-coordinates.
- Over time, all countries have experienced increases in their GDPs (and, as per the bullet point above, increases in their average life expectancies).

Let's break down this plot a bit more.

Question 2

What real-world object does each point (bubble) represent? **Hint:** what happens when you hover your cursor over each bubble?

Solution: Each point represents a **country**.

Question 3

Why are some points bigger than others? Why are the sizes of the points changing as the video progresses?

Solution: The size of each point corresponds to its population, as indicated by one of the options on the website. As such, some points are bigger because they correspond to countries with larger populations.

Now, let's personalize a bit! That is, it's time for you to pick a country you're interested in to examine further.

Question 4

Select a country from the countries listed to the right side of the page and play the video again. Do you notice anything surprising in the country's trajectory regarding life expectancy? Can you justify any of these "surprises" based on country's history? Explain.

Solution: Again, answers will vary! I chose the United States, since we are all currently *in* the US. When it comes to life expectancies, we see a few things:

- Between 1800 and around 1860, the life expectancy stays constant even though the GDP is increasing (along with the population).
- In the 1860s, there is a dip in life expectancies. Perhaps not coincidentally, the **US Civil War** took place between 1861 and 1865, which is likely what is driving the observed decrease in life expectancies.
- We see a period of steady incline until around the 1910s, when we observe another dip in life expectancies likely stemming from **World War I** (indeed, *all* points exhibit a drop in their y-coordinates during this period).
- Finally, there is yet another period of steady increase in life expectancies until a slight dip in 2020, likely corresponding to the COVID-19 pandemic

One thing to note: this question (hopefully) highlights the importance of background knowledge when it comes to interpreting plots. Statisticians and data scientists use the term **domain knowledge** to refer to any background knowledge needed to extract additional information from plots/analyses - indeed, statisticians and data scientsts are often *highly* reliant on domain knowledge (which is one of the reasons we like to collaborate with people from a wide variety of different disciplines!)

Statistical Terminology

Alright, let's start making things a little more formal. First, recall the following two pieces of vocabulary:

- **Observational Units** are the objects on which data is taken. These can be things like people, households, countries, cats - you name it!

- A **variable** is an attribute of an observational unit that we can measure. Note that the value of a variable can vary from unit to unit - for example, we expect different people to have different heights.

Question 5
Describe the observational unit (i.e. identify what the observational units are).

Solution: A single observational unit is a **country**, since the data measures various attributes (GDP, Life Expectancy, etc.) on countries.

Note: On exams, we'll typically ask you to describe the observational unit in singular form (e.g. instead of saying "countries," say "country"). On quizzes it won't matter too much (since quizzes will be multiple choice), but I wanted to convey this information to everyone.

Question 6
What are the five variables represented in the plots?

Solution: The five variables are:

- GDP (Gross Domestic Product) per capita
- Life Expectancy
- Population
- Country name (which is visible after hovering over each point)
- Global region (which is indicated by the color of each point)

Note: as someone pointed out in lecture, there are actually *six* variables represented - **Year** is another one! Admittedly, Year is a bit of a tricky one because for each static plot (i.e. for each snapshot in time) there is only one value for the Year, however if we consider the plots in series then Year is definitely a variable whose information is encoded in the plots.

Additionally, recall that variables come in (broadly speaking) two types:

- **Categorical variables** are variables whose values are "words-" more specifically, whose values fall into a fixed set of "categories" (which we often call **levels**).
- **Quantitative variables** are variables whose values are numbers, and those such that it makes interpretive sense to perform algebraic manipulations on the values.

⚠ Caution

Just because a variable has values that are numbers doesn't mean it's a quantitative one! For example, consider a dataset in which observational units are people and one variable is **Birth Month**. We often write birth months using numbers - e.g. "1" for January, "2" for February, etc. This doesn't make **Birth Month** a quantitative variable - that's because "1" + "2" really means "January" + "February," which has no interpretive meaning.

In more technical statistical speak: categorical variable sometimes have values that are **encoded** as numbers.

Question 7

For each of the variables you identified in Question (6) above, classify them as either categorical or quantitative.

Solution:

- GDP is **quantitative** - its values are numbers, and there is interpretive meaning behind performing arithmetic operations on its values.
- Life Expectancy is **quantitative** - its values are numbers, and there is interpretive meaning behind performing arithmetic operations on its values.
- Population is **quantitative** - its values are numbers, and there is interpretive meaning behind performing arithmetic operations on its values.
- Country name is **categorical** - its values are "text," with each observation falling into one of a fixed set of levels.
- Global region (which is indicated by the color of each point) is **categorical** - its values are "text," with each observation falling into one of a fixed set of levels.
- Year is a **quantitative** variable - its values are numbers, and there is interpretive meaning behind performing arithmetic operations on its values (2020 + 2021 = 4041).

Going Beyond

If you're curious, there's actually a second level by which we can categorize variables (this is outside the scope of this course, but, I wanted to put this here for those of you who are curious).

Consider a variable that tracks the number of bike accidents along a particular stretch of road. This is a quantitative variable - its values are numbers, and there is interpretive meaning behind arithmetic operations on its values (1 accident + 2 accidents = 3 accidents). However, there seems to be this extra constraint: namely that we can't have a fractional number of accidents!

This is what leads us into the second level of classification: quantitative variables can be further subdivided into **Discrete** and **Continuous**. Discrete variables are numerical variables whose possible values "have constraints," and continuous variables are quantitative variables whose possible values "do not have constraints." So, "Number of Accidents" example of a discrete variable, because the possible values are *constrained* to be integers. **An Aside:** see if you can go back to the GDP/Life Expectancy data and classify the quantiative variables as either discrete or continuous.

But, to stress, both discrete and continuous variables are quantitative variables. When checking whether a variable is quantitative or categorical (which is definitely fair game for quizzes etc.), you should do the check we did in lecture; namely, ask "what does this value actually represent - is it really a number, or is the number just a placeholder for some other non-numeric value?"

Part 2: Monetary Allocation of the Brazilian Government

We'll now take a look at some information published by the Brazilian government on expenditure allocation; specifically, we'll be focusing on a plot posted to their website that displays the five sub-categories within Hospital and Outpatient Care sub-area that received the most resources (presumably money) in 2021. A quick disclaimer: the original plots have text in Brazilian Portuguese, but we will be exploring the plots as translated to English by Google Chrome's auto-translation feature.

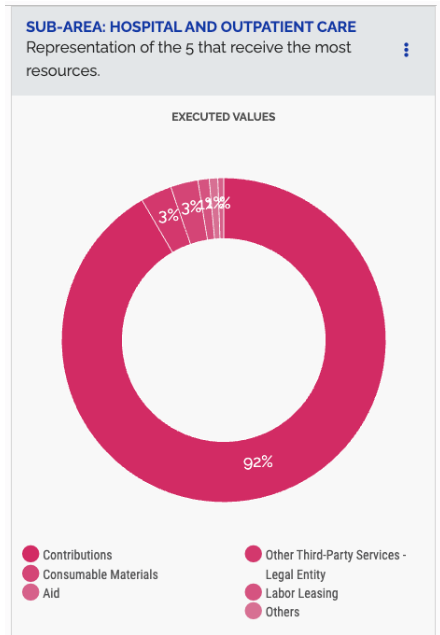

Figure 1: A donut chart of allocation percentages to the hospital and outpatient care sub-area of the Brazilian Government, in 2021. Image source: `https://portaldatransparencia.gov.br/funcoes/10-saude?ano=2021`

Question 8

What's the "story" this plot tells us? (This is an intentionally open-ended question!)

Solution: Answers may vary! Generally speaking, there are a few things we can see from the plot:

- A very high percentange (92%) of resources were allocated to the "Contributions" category
- Very few resources were allocated to the remaining categories, in roughly equal proportions.

Another valuable skill to have is the ability to make constructive criticism of plots!

Question 9

Was Figure (1) easy to read? If not, what specific aspects were difficult to read? How did those aspects impact your ability to identify the "story" the plot is trying to tell?

Solution: Answers may vary, but I think most people would agree that Figure (1) is very difficult to read! Specifically, in choosing to color the categories with colors that are visually so similar, the story of the plot is largely lost. For example: I actually "cheated" a bit in the previous question - I had to go to the original website to see that the 92% was allocated to the "Contributions" category! Admittedly, the plot online is interactive, however it is important to think about making plots that are highly adaptable to different media modes - something the plot authors of the plot in Figure (1) likely did not take into consideration...

Part 3: US Living Situations

Below is a figure demonstrating how the different living situations of US households (living alone, Married with no Kids, etc.) has changed between 1980 and 2025.

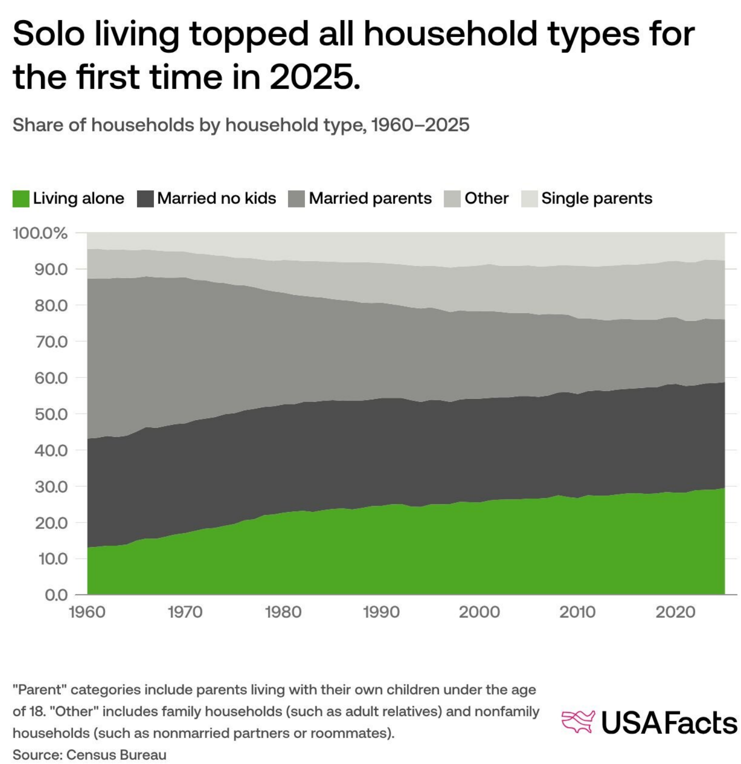

Figure 2: A ribbon plot demonstrating how US households' living situations have changed over time. Categories include: Living Alone, Married No Kids, Married Parents, Other, and Single Parents. Image Source: `https://usafacts.org/articles/semiquincentennial-snapshot-the-us-at-250-years/`

Question 10

Describe the observational unit.

Solution: The observational units are **households**, since information was collected on each *household's* living situation.

Question 11

What type of variable (categorical or quantitative) is **Household Type**? If it is categorical, what are its levels?

Solution: **Household Type** is a categorical variable, with levels: Living Alone, Married No Kids, Married Parents, Other, and Single Parents.

Let's practice extracting some more detailed information from plots.

Question 12

Approximately what percentage of households in 1960 were in the **Living Alone** category? How can you tell (i.e. what parts of the plot did you use to come to your answer)?

Solution: It appears as though the percentage of households in the **Living Alone** category in 1960 was **13%**, since the "Living Alone" category had a height of 13.0 in 1960.

Question 13

Approximately what percentage of households in 2020 were in the **Living Alone** category? How can you tell (i.e. what parts of the plot did you use to come to your answer)?

Solution: It appears as though the percentage of households in the **Living Alone** category in 2020 was around **29%**, since the "Living Alone" category had a height of around 29.0 in 2020.

Question 14

Approximately what percentage of households in 1960 were in the **Married Parents** category? How can you tell (i.e. what parts of the plot did you use to come to your answer)?

Solution: It appears as though the percentage of households in the **Married Parents** category in 1960 was 87% - 42% = **45%**, which represents the width of the "Married Parents" band on the plot in 1960. The key is to note that the percentages within each category are not just the raw y-axis values, but rather the *widths* of the category bands.